

Methods of proof – 5

Non-constructive existence: counting, diagonal, category,
chance

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Theoretical minimum.

Fact 1 (counting as existence). If the objects to be avoided form a countable family and the candidates form a continuum, a good candidate exists. This is the cheapest existence proof in analysis: it produces transcendental numbers, points avoiding a countable set of obstacles, and functions whose set of singularities is prescribed.

Theorem 1 (the diagonal argument). Cantor’s diagonal shows that $\mathbb{R}$ is uncountable, that the power set is larger than the set, and — in its “moving” form — extracts from a sequence of sequences one that differs from each of them; the same construction extracts a subsequence converging simultaneously in countably many senses, which is how Arzelà–Ascoli and Helly’s selection theorem are proved.

Theorem 2 (Baire as a machine). In a complete metric space a countable union of nowhere dense sets has empty interior, so a residual set is non-empty. This produces a continuous nowhere differentiable function (Differentiable functions – 8), a continuous function whose Fourier series diverges at a point (Fourier analysis – 1) and a great many objects nobody can write down.

Fact 2 (chance versus category). The probabilistic method proves existence by showing that a random object works with positive probability (Probability – 4, Inequalities – 8). It and Baire’s theorem are two different notions of “typical”: a set may be residual and of measure zero at the same time, so the two methods can give opposite answers about what a typical object looks like.

1. Prove that the algebraic numbers are countable and deduce the existence of transcendental numbers; then construct a concrete transcendental number by Liouville's criterion and explain what the counting argument does not give.
2. (IMS 2006) Let X be a separable metric space and let $f: X \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow a} f(x)$ exists for every $a \in X$. Prove that the set of points at which f is not continuous is countable. (Two further counting problems of this kind — an infinite set all of whose finite subsums are bounded, and a family of spheres meeting pairwise in at most one point — are in the homework of Topology of the real line – 1.)
3. (IMC) Let $f: (0, 1) \rightarrow [0, \infty)$ vanish except at distinct points $a_1, a_2, \dots$, and put $b_n = f(a_n)$.
 - a) Prove that if $\sum b_n < \infty$, then f is differentiable at at least one point of $(0, 1)$.
 - b) Prove that for any non-negative (b_n) with $\sum b_n = \infty$ there is a sequence (a_n) for which the function f so defined is nowhere differentiable.
4. Prove by the diagonal argument that $[0, 1]$ is uncountable, that no sequence of functions contains all continuous functions, and extract from a uniformly bounded sequence of functions on $\mathbb{Q} \cap [0, 1]$ a pointwise convergent subsequence.
5. Prove that the set of continuous functions on $[0, 1]$ that are monotone on some subinterval is of first category, and deduce the existence of a continuous nowhere monotone function; then compare with the probabilistic construction through the Wiener process.